

The local $\mathbb{Z}_2$ marker under periodic boundary conditions: per-site frames without a per-site cost

Abstract

Evaluating the local marker at each site in its own coordinate frame — so that the site sits as far as possible from the branch cut of the periodic position operator — appears to require rebuilding the operators X , Y and M inside the site loop, at a cost of $O(N)$ full matrix triple products. We show that it does not. Three observations reduce the whole calculation to a *single* $D \times D$ matrix product for all sites at once, i.e. to the same cost as the present global-frame implementation: (i) the position operators are block-constant, so the geometry factorises out of the trace as a real scalar weight; (ii) only the diagonal blocks of the triple product are ever read; and (iii) the frame label coincides with the outer index of that product, so N different frames can be stored as an elementwise rescaling of $\tilde{S}$ rather than applied one at a time. A fourth observation — that hermiticity of $\tilde{S}$ makes the weight's partner term redundant — removes a further factor of two. We also record that the global origin is exactly irrelevant when $\tilde{S}^2 = I$, and that centring on $\lfloor L/2 \rfloor$ is the minimum-image convention only for even L .

1 Setting and notation

Let the sample have N sites with 4 orbitals each, so the single-particle Hilbert space has dimension $D = 4N$. A basis state is labelled by a site $a \in \{0, \dots, N-1\}$ and an orbital $\alpha \in \{0, 1, 2, 3\}$, flattened as $4a + \alpha$ (orbital fast, site slow — the convention produced by `np.repeat(·, 4)`). For a $D \times D$ matrix A we write A_{ab} for the 4×4 block

$$(A_{ab})_{\alpha\beta} = A[4a + \alpha, 4b + \beta], \quad (1)$$

and Π_a for the orthogonal projector onto the four orbitals of site a , so that $\sum_a \Pi_a = I_D$.

The auxiliary operator $\tilde{S}$ is Hermitian and, after the sign flattening, involutive:

$$\tilde{S}^\dagger = \tilde{S}, \quad \tilde{S}^2 = I_D. \quad (2)$$

Both properties are used below and both are guaranteed by construction in `S_tilde_one_parameter`.

Given site coordinates x_a, y_a , define the diagonal position operators $X = \text{diag}(x_a \text{ repeated } 4\times)$ and likewise Y . The marker implemented in `modules/marker.py` is

$$M = \tilde{S}X\tilde{S}Y\tilde{S} - \tilde{S}Y\tilde{S}X\tilde{S}, \quad \nu_i = -\frac{\pi}{8} \text{Tr Im } M_{ii}, \quad (3)$$

where M_{ii} is the 4×4 diagonal block at site i and Tr runs over the four orbitals only.

2 Two elementary lemmas

Lemma 1 (Block-constancy of the position operator). $X = \sum_a x_a \Pi_a$, and consequently, for any $D \times D$ matrix A ,

$$[AX]_{ab} = x_b A_{ab}, \quad [XA]_{ab} = x_a A_{ab}. \quad (4)$$

Proof. X assigns the same number x_a to all four orbitals of site a , which is precisely the statement $X = \sum_a x_a \Pi_a$. Then $[AX]_{ab} = \sum_c A_{ac}(X)_{cb} = A_{ab} x_b$, since $(X)_{cb} = \delta_{cb} x_b I_4$. $\square$

The content of Lemma 1 is that a position operator acts on the block structure as a *scalar*, never mixing orbitals. This is what lets the geometry leave the trace.

Lemma 2 (The global origin is irrelevant). *If $\tilde{S}^2 = I$, then M_{ii} is unchanged under a uniform translation $x_a \mapsto x_a + c$, $y_a \mapsto y_a + d$ for all a , with c, d arbitrary constants.*

Proof. Write $F(u, v) = \tilde{S} U \tilde{S} V \tilde{S}$ for the triple product with diagonal insertions built from vectors u, v , so that $M = F(x, y) - F(y, x)$, and note that F is bilinear in (u, v) . Under the translation,

$$F(x + c\mathbf{1}, y + d\mathbf{1}) = F(x, y) + dF(x, \mathbf{1}) + cF(\mathbf{1}, y) + cdF(\mathbf{1}, \mathbf{1}). \quad (5)$$

Setting one argument to $\mathbf{1}$ replaces an insertion by I_D , and $\tilde{S}^2 = I$ then collapses the product. Using Lemma 1,

$$F(x, \mathbf{1})_{ii} = [\tilde{S}X\tilde{S}\tilde{S}]_{ii} = [\tilde{S}X]_{ii} = x_i\tilde{S}_{ii}, \quad F(\mathbf{1}, x)_{ii} = [\tilde{S}\tilde{S}X\tilde{S}]_{ii} = [X\tilde{S}]_{ii} = x_i\tilde{S}_{ii}, \quad (6)$$

so $F(x, \mathbf{1})_{ii} = F(\mathbf{1}, x)_{ii}$ and likewise for y . Every correction therefore cancels between the two orderings in $M = F(x, y) - F(y, x)$: the d -linear terms cancel, the c -linear terms cancel, and the cd terms cancel identically. Hence M_{ii} is unchanged. $\square$

Remark 1. Lemma 2 says the recentring $x \mapsto x - \frac{1}{2}(N_x - 1)$ in the present code has no effect whatsoever on the output (verified numerically at the level of 10^{-15} ; see Table 1). Under open boundaries there is therefore nothing to choose. Under *periodic* boundaries the coordinate is only defined modulo L , and what a per-site frame really selects is not an origin but the location of the *branch cut* of that coordinate — which is not a uniform translation, and which Lemma 2 does not protect. That is the whole reason the per-site frame matters at all.

3 The pair expansion

Proposition 1 (Pair expansion of the diagonal block). *With $w_{ab} = x_a y_b - y_a x_b$,*

$$M_{ii} = \sum_{a,b} w_{ab} \tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}, \quad \nu_i = -\frac{\pi}{8} \sum_{a,b} w_{ab} \operatorname{Im} \operatorname{Tr} [\tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}]. \quad (7)$$

Proof. Read the five-factor product in blocks. A block matrix product resolves as $(ABC)_{ii} = \sum_{a,b} A_{ia} B_{ab} C_{bi}$, the sum running over sites. Grouping (3) as $(\tilde{S}X)(\tilde{S}Y)(\tilde{S})$ and applying Lemma 1 to each bracket,

$$[\tilde{S}X\tilde{S}Y\tilde{S}]_{ii} = \sum_{a,b} [\tilde{S}X]_{ia} [\tilde{S}Y]_{ab} \tilde{S}_{bi} = \sum_{a,b} x_a y_b \tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}, \quad (8)$$

because x_a and y_b are numbers and commute out of the 4×4 products. The second term of (3) is the same expression with $x \leftrightarrow y$, giving $\sum_{a,b} y_a x_b \tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}$. Subtracting yields (7); taking $\operatorname{Tr} \operatorname{Im}$ and using that w_{ab} is real completes the statement. $\square$

Equation (7) is the structural heart of the matter. The marker at site i is a sum over *pairs* of sites of

- a real scalar weight w_{ab} , which carries all of the geometry, and
- a 4×4 matrix factor $\tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}$, which carries all of the quantum mechanics and knows nothing about positions.

The weight is antisymmetric, $w_{ab} = -w_{ba}$: it is the wedge product $\mathbf{r}_a \wedge \mathbf{r}_b$.

Proposition 2 (Per-site frames). *Let each site i carry its own coordinate frame, in which site a has coordinates D_{ia}^x, D_{ia}^y . Then*

$$\nu_i = -\frac{\pi}{8} \sum_{a,b} \left(D_{ia}^x D_{ib}^y - D_{ia}^y D_{ib}^x \right) \text{Im Tr} \left[\tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi} \right]. \quad (9)$$

Proof. The proof of Proposition 1 used only that the position insertion for the frame of site i is diagonal and block-constant, i.e. Lemma 1. Each site's own frame has that property, so the derivation applies verbatim with $x_a \mapsto D_{ia}^x$ and $y_b \mapsto D_{ib}^y$. $\square$

So D^x and D^y are $N \times N$ tables whose i -th row holds the coordinates as seen from site i . Nothing has been approximated.

4 Packing the frames into a single matrix product

Formula (9) looks like N separate double sums. It is not: the frame label i enters only through the *outer* indices of the product, and that is exactly what lets it be absorbed into the matrices.

Proposition 3 (Frame packing). *Let J_4 be the 4×4 matrix of ones, $\odot$ the elementwise (Hadamard) product, and define the two $D \times D$ matrices*

$$P = (D^x \otimes J_4) \odot \tilde{S}, \quad Q = ((D^y)^\top \otimes J_4) \odot \tilde{S}, \quad (10)$$

that is, blockwise $P_{ia} = D_{ia}^x \tilde{S}_{ia}$ and $Q_{bi} = D_{ib}^y \tilde{S}_{bi}$. Then, for every site i ,

$$\sum_{a,b} D_{ia}^x D_{ib}^y \tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi} = [P \tilde{S} Q]_{ii}. \quad (11)$$

Proof. By the block resolution of a triple product, $[P \tilde{S} Q]_{ii} = \sum_{a,b} P_{ia} \tilde{S}_{ab} Q_{bi}$. Substituting the definitions of P and Q gives $\sum_{a,b} D_{ia}^x \tilde{S}_{ia} \tilde{S}_{ab} D_{ib}^y \tilde{S}_{bi}$, and the scalars commute out. $\square$

The transpose in Q is not decoration: the table entry is indexed $[i, b]$ while the matrix element sits at row b , column i .

Remark 2 (Why this is the crucial step). In $[P \tilde{S} Q]_{ii}$ the row index of P and the column index of Q are *both* i . Any dependence on i living on those indices is therefore precisely what a diagonal-block extraction sees. “Site i is evaluated in frame i ” is thus the same statement as “row i of P is scaled by frame i ”: the frames become *matrix-shaped data* instead of a per-site operation. Compare with the global-frame code, where $P = X \tilde{S}$ rescales rows by one common vector and every row shares a frame. The matrix has the same shape and the product the same cost — it now simply carries N distinct frames instead of one.

Proposition 4 (Only one of the two terms is needed). *Because $\tilde{S}$ is Hermitian, $T_{iab} \equiv \text{Im Tr}[\tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}]$ is antisymmetric under $a \leftrightarrow b$, and consequently*

$$\nu_i = -\frac{\pi}{4} \sum_{a,b} D_{ia}^x D_{ib}^y \text{Im Tr} [\tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}] = -\frac{\pi}{4} \text{Im Tr} [P \tilde{S} Q]_{ii}. \quad (12)$$

Proof. Hermiticity gives $\tilde{S}_{ba} = \tilde{S}_{ab}^\dagger$, so

$$\text{Tr}[\tilde{S}_{ib} \tilde{S}_{ba} \tilde{S}_{ai}] = \text{Tr}[\tilde{S}_{bi}^\dagger \tilde{S}_{ab}^\dagger \tilde{S}_{ia}^\dagger] = \text{Tr}[(\tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi})^\dagger] = \overline{\text{Tr}[\tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}]}, \quad (13)$$

whence $T_{iba} = -T_{iab}$. Relabelling $a \leftrightarrow b$ in the second half of $\sum_{a,b} (D_{ia}^x D_{ib}^y - D_{ia}^y D_{ib}^x) T_{iab}$ shows that it equals the first half, so the antisymmetrised sum is twice the plain one. $\square$

Corollary 1 (Algorithm and cost). *All N site markers follow from one $D \times D$ matrix product:*

$$R = P \tilde{S} \quad (\text{the only } O(D^3) \text{ step}), \quad \nu_i = -\frac{\pi}{4} \operatorname{Im} \sum_b \operatorname{Tr}[R_{ib} Q_{bi}]. \quad (14)$$

The remaining contraction touches only the N diagonal blocks, at cost $O(N^2 \cdot 4^3)$, which is negligible. In `numpy`, with p, q the orbital indices:

```
DX = np.repeat(np.repeat(Dx, 4, axis=0), 4, axis=1) # once per lattice
DY = np.repeat(np.repeat(Dy, 4, axis=0), 4, axis=1)
P, Q = DX * S, DY.T * S
R = (P @ S).reshape(N, 4, N, 4)
marker = -(pi / 4) * np.imag(np.einsum('ipbq,bqip->i', R, Q.reshape(N, 4, N, 4)))
```

Rebuilding M inside the site loop costs $O(ND^3) = O(64N^4)$; the above costs $O(D^3) = O(64N^3)$. The saving is a factor $\sim D = 4N$.

5 Choosing the frames: minimum image and the parity of L

It remains to fix D^x, D^y . Shifting the sample so that site i lands at the box centre and then reducing modulo L sends site a to $[(x_a - x_i) + L/2] \bmod L$, i.e. to

$$D_{ia}^x = \operatorname{mi}(x_a - x_i) \equiv \left[(x_a - x_i) + \frac{L}{2} \right] \bmod L - \frac{L}{2}, \quad (15)$$

up to a uniform additive constant, which by Lemma 2 is immaterial. So the per-site frame is nothing but the **minimum-image convention** for the relative vectors $\mathbf{r}_a - \mathbf{r}_i$, and (15) is a two-line vectorised precomputation, done once per lattice, entirely outside the site loop. (The same minimum-image logic already appears in `displacement2D.kwant`.)

Remark 3 (Centring on $\lfloor L/2 \rfloor$ is minimum image only for even L). Placing site i at $\lfloor L/2 \rfloor$ rather than at $L/2$ displaces it by half a lattice spacing when L is odd. The branch cut then sits at relative distance $(L+1)/2$ on one side and $(L-1)/2$ on the other, instead of symmetrically at $\pm L/2$. For a crystalline sample the coordinates are integers, no site ever falls in the disputed half-spacing window, and the two conventions agree exactly. With Gaussian jitter some sites do fall in it and are assigned to opposite images, which changes the marker by an amount that decays with L (Table 3). Minimum image is therefore not merely an equivalent reformulation: being symmetric by construction, it is the faithful implementation of “as far from the cut as possible”, and it removes an $O(\xi/L)$ artefact that the floor-centred frame carries at odd L .

Remark 4 (Locality, and why the frame matters). $\tilde{S}$ is the sign of a gapped operator and is exponentially localised, so T_{iab} is negligible unless both a and b lie within a few localisation lengths ξ of site i . The per-site frame pushes the branch cut a distance $\sim L/2$ away from site i , so the error it commits is exponentially small in $L/2\xi$. In the global frame, by contrast, sites near the cut have it running straight through their correlation neighbourhood, which is why the unshifted periodic marker is meaningless (Table 1). Since ξ grows as the $\tilde{S}$ gap $\Delta_{\tilde{S}}$ closes, the scheme is controlled by $\Delta_{\tilde{S}}$ and L jointly.

6 Numerical verification

All checks below were run through the existing pipeline (`AmorphousLattice_2d` $\rightarrow$ `S0_syst_kwant` $\rightarrow$ `spectrum` $\rightarrow$ `S_tilde_one_parameter`), with $M = -2$, $A = 1$, $\lambda_R = 1$, $\lambda_D = 0$, $W = 0$, periodic boundaries unless stated.

Table 1: Residuals of each step of the derivation, against the existing `local_marker` or against a literal per-site- M loop.

Claim	Check	Residual
Lemma 2	$\max_i \nu_i(x) - \nu_i(x + 3.7) $	2.0×10^{-15}
Lemma 2	$\max_i \nu_i(x) - \nu_i(x + 100) $	5.5×10^{-13}
Prop. 1	$\ M_{ii} - \sum_{ab} w_{ab} \tilde{S}_{ia} \tilde{S}_{ab} \tilde{S}_{bi}\ _{\max}$	2.3×10^{-15}
Prop. 3	$\ M_{ii} - [P\tilde{S}Q]_{ii} + [P'\tilde{S}Q']_{ii}\ _{\max}$	9.0×10^{-16}
Cor. 1	$\max_i \text{Tr}[P\tilde{S}Q]_{ii} - \text{einsum} $	2.4×10^{-15}
Prop. 4	$\max_i \text{two-term} - \text{one-term doubled} $	3.8×10^{-15}
Prop. 2	$\max_i \text{per-site-}M \text{ loop} - \text{table formulation} $	7.8×10^{-16}
<i>Consistency of the scheme itself</i>		
Uniformity	crystalline PBC: $\max_i \nu_i - \min_i \nu_i$	$\leq 10^{-14}$
Necessity	global-frame PBC: $\max_i \nu_i - \min_i \nu_i$	5.09

Table 2: Cost of one marker evaluation: rebuilding M per site versus Corollary 1 (two-term version timed; the one-term version of Proposition 4 halves it again). Amorphous, $w = 0.1$.

L	N	D	tables	per-site loop	speed-up
10	100	400	0.009 s	1.5 s	167×
12	144	576	0.025 s	6.1 s	248×
15	225	900	0.076 s	29 s	382×
18	324	1296	0.19 s	110 s	570×

Table 3: Floor-centred frames versus minimum-image frames, $\max_i |\nu_i^{\text{floor}} - \nu_i^{\text{mi}}|$, illustrating Remark 3. The two agree identically for even L and for any crystalline sample; they differ for odd L with amorphicity, by an amount decaying with L .

L (odd)	5	7	9	11	13
amorphous $w = 0.12$	0.48	0.12	0.050	0.023	0.022

For every even L tested (6, 8, 10, 12, 14), and for *every* L of either parity in the crystalline case, the two conventions agree to $\lesssim 2 \times 10^{-14}$, i.e. to machine precision.

Table 4: Convergence of the periodic all-site average $\langle \nu \rangle$ with system size, at the clean optimal angle. Under minimum-image frames every site of a crystalline periodic sample gives the same value, as translational symmetry demands.

L	6	8	10	12	14	16
crystalline, $\langle \nu \rangle$	-0.898	-0.992	-0.997	-0.998	-0.997	-0.9996
amorphous $w = 0.15$, $\langle \nu \rangle$	-0.864	-0.958	-0.974	-0.956	-0.974	—
amorphous $w = 0.15$, site std.	0.150	0.162	0.165	0.164	0.163	—

7 Summary

1. The global origin is exactly irrelevant (Lemma 2); under periodic boundaries a per-site frame selects the *branch cut*, and that choice is the minimum-image convention (15).

2. The geometry separates from the trace as a real antisymmetric weight (Proposition 1), and per-site frames enter it for free (Proposition 2).
3. Because the frame label is the outer index of the triple product, N frames can be packed into an elementwise rescaling of $\tilde{S}$ (Proposition 3); since only diagonal blocks are read, all sites follow from one $D \times D$ product (Corollary 1), and hermiticity halves even that (Proposition 4).
4. Net effect: periodic, per-site-framed, all-site-averaged markers at *lower* cost than the current global-frame open-boundary implementation.